ISSS603 Applied Data Science for Customer Insights

Mid-Term Report

Group 1

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1. Introduction

As part of our analysis, we were provided with dataset highlighting various operation aspects of LushProtein from December 2019 to March 2026; namely on customer transactions, product offerings, discounts, campaigns and subscription data. In this report, we will discuss on our data pipeline as well as the analytical insights gathered.

1.1 Temporal Coverage

Period of analysis: 2019-12-31 to 2026-03-30

Period	Orders	Unique Customers	Notes
2019	3	3	Incomplete year — excluded from cohort analysis
2020	2,847	1,696	Full year, zero discounting
2021	6,259	3,815	Peak revenue year (S$848K)
2022	3,710	2,299	First discounting introduced
2023	2,253	1,399	Revenue collapse year
2024	4,261	2,621	Recovery, heavy discounting
2025	6,407	4,107	Highest order volume, continued discounting
2026 (Partial)	1,610	1,230	Jan–Mar only; excluded from full-year analysis

1.2 Store and currency coverage

Store	Order	Revenue (Own Currency)	FX Rate	Revenue (SGD equivalent)
SG	16,041	SGD $1,913,068	1.00	S$1,913,068
MY	11,309	RM $3,958,566	3.30	S$1,199,566
HK	2	HK$1,943	6.10	S$319
All markets	**27,350**			**S$3,112,952**

Notes: The three stores use different currencies. The raw data contains no FX rates.

Mitigation applied (May 2026): 5-year average exchange rates (2020–2026) applied at data load time:

- 1 SGD = 3.30 MYR

- 1 SGD = 6.10 HKD

These are 5-year average rates (not spot rates at each transaction date). This introduces a measurement error in absolute revenue figures of up to ±10%. The direction and relative magnitude of all findings are unaffected — retention, repeat rates, and channel comparisons are count-based and FX-neutral.

Known limitations of fixed-rate FX approach:

1. SGD/MYR moved between ~3.0 and ~3.5 over 2020–2026. Early-year MY revenue (2020–2021) may be understated/overstated by up to 10%.

2. MY market was effectively dormant by 2025 (only S$9,617 SGD in 2025 vs S$441,022 in 2021), so the combined totals are dominated by SG data in recent years.

3. HK store (2 orders, S$319 SGD equivalent) is immaterial.

2. Summary Statistics and Distributions

2.1 Order Revenue Distribution (All Markets – in SGD after FX conversion)

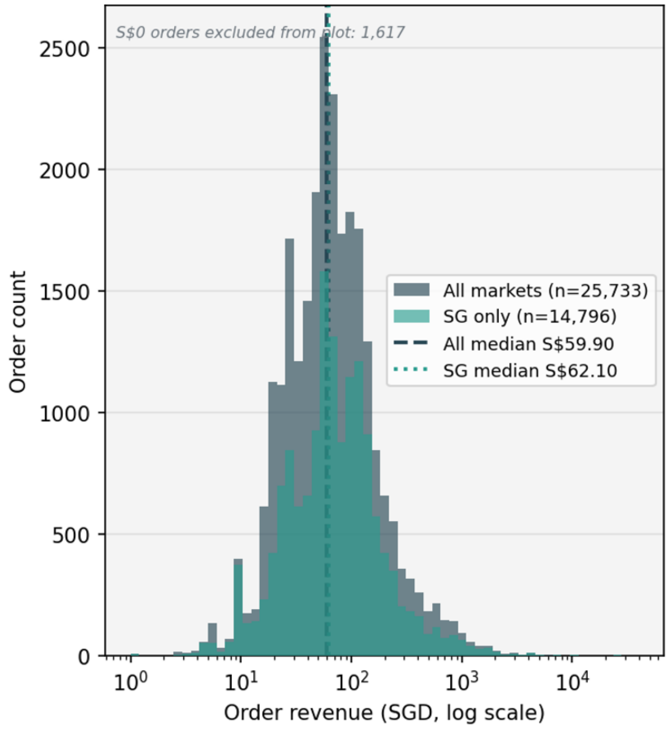


Figure 2.1a - Order Revenue Distribution
(positive-revenue orders only; log x-axis)

Statistics	**All markets (SGD)**	**SG only (SGD)**	**MY only (SGD equiv.)**
Count	27,530	16,039	11,309
Mean	\$113.83	\$119.28	\$106.07
Median	\$59.90	\$62.10	\$55.48
Standard deviation	\$413.76	\$405.16	\$425.60
25th percentile	\$29.40	\$29.00	\$29.70
75th percentile	\$108.41	\$115.00	\$96.67
90th percentile	\$202.48	\$206.66	-
Maximum	\$34,618	S\$26,520	\$34,618

The mean (S\$114) is nearly double the median (S\$60) across all markets, indicating a strongly right-skewed distribution driven by a small number of high-value wholesale/reseller orders. This is why per-customer LTV is reported as a median in key comparisons (refer to section 3.4 – Outlier / Wholesale Orders).

Reason for the all-markets median (S\$59.90) is lower than SG median (S\$62.10):MY orders have a slightly lower median (S\$55.48 SGD-equivalent) due to smaller average pack sizes and the fixed-rate FX approximation. Combining both markets pulls the combined median below the SG-only figure.

The largest single order in the dataset is a MY wholesale order of MYR 114,240 (≈ S\$34,618 after FX conversion), which exceeds the largest SG order of S\$26,520, leading to the all-markets maximum value (\$34,618) is higher than the maximum value in SG market (\$26,520)

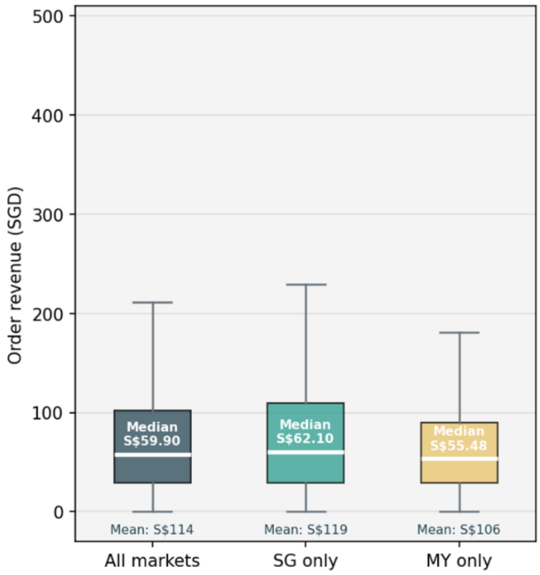


Figure 2.1b – Revenue Box Plot by Market
(capped at S$500; outliers hidden)

2.2 Order per Customer Distribution

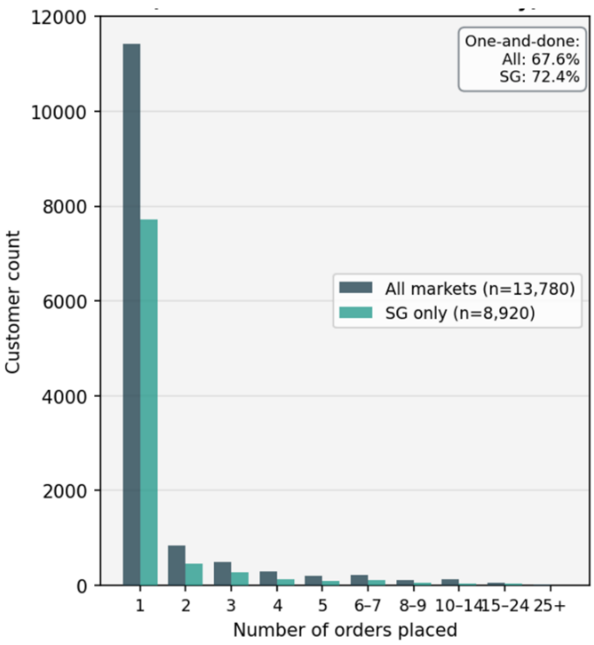


Figure 2.2 - Order per Customer Distribution
(All markets combined vs SG only)

Statistics	**All markets**	**SG market only**
Mean	1.98	1.8
Median	1	1
75th percentile	2	2
90th percentile	4	3
Maximum	667	667
# of unique customers	13,780	8,920
# of customers with exactly 1 order	9,321 (67.6%)	6,454 (72.4%)
# of customers with >10 orders	268 (1.9%)	111 (1.2%)
# of customers with >20 orders	35 (0.3%)	-

67.6% of all-markets customers — and 72.4% of SG-only customers — made exactly one purchase. The SG one-and-done rate is slightly higher, partly because many MY customers who made multiple purchases across the more established MY market have longer order histories. The extreme maximum of 667 orders belongs to a single customer and is almost certainly a reseller or test account.

2.3 Discount distribution (All discounted orders)

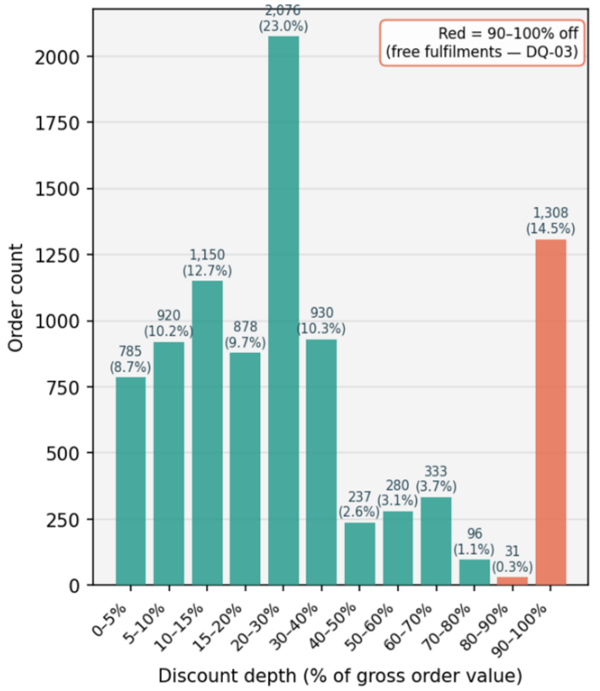


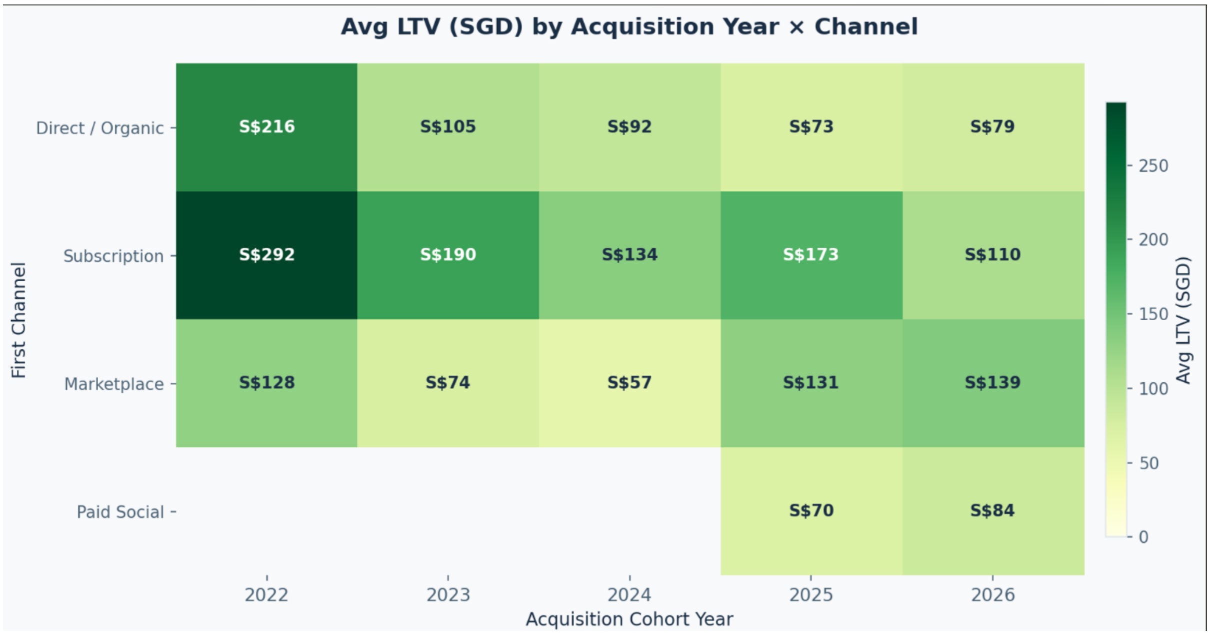
Figure 2.4 Discount distribution – All markets
(9,024 orders with any discount; 33% of total)

Discount depth = $\frac{Discount\ Price}{Total\ Price+Discount\ Price}$

(discount as a percentage of the gross (pre-discount) order value)

The 14.5% spike at 90–100% off (1,308 orders where the customer paid nothing or near-nothing) is a notable anomaly — these are real product shipments (referral rewards, PR samples, subscription welcome gifts) at zero customer cost. (refer to 3.3 – 100% Discount Orders).

2.4 LTV analysis by acquisition year and channel



Subscription-acquired customers consistently generate the highest lifetime value (LTV) across most cohorts, peaking at S$292 in 2022, indicating that subscription remains the most valuable acquisition channel despite a general decline in LTV for newer cohorts.

However, customer quality appears to be declining for several traditional acquisition channels in newer cohorts. Marketplace LTV is mixed by year (e.g. 2023 avg LTV S$74 vs Direct/Organic S$105; 2025 Marketplace S$131 vs Direct S$73 on the full midterm base) and must be interpreted with caution because of Shopify tracking limits (see §3.8): marketplace repeat rate (14.4%) remains well below Direct/Organic (33.5%), and off-platform behaviour is undercounted. We do not recommend increased marketplace spend on LTV grounds alone.

2.5 Missing Value Summary

Column	Null Count	% Missing	Notes
Tags	17,026	62.3%	Expected behaviour.
Browser: UTM Source	25,945	94.9%	Most orders lack UTM tracking (refer to 3.5 – Missing Values in UTM Attributions))
Browser: UTM Medium	25,945	94.9%	Same as above
Browser: UTM Campaign	25,978	95.0%	Same as above
Browser: Referrer Domain	22,383	81.8%	Expected for direct/organic traffic
Line: Product Handle	13,484	49.3%	Order-header rows without line items (refer to 3.6 – Line-Item vs Order-Level Row Structure))
Shipping: Country	1,055	3.9%	Some orders missing shipping destination
Order Fulfilment Status	207	0.8%	Minor gap
second_order_date	9,321	67.6%	Expected: 67.6% of customers are one-time buyers

3. Data Issues and Discrepancies

3.1 Mixed Currencies

The orders in the dataset cover 3 different countries (Singapore, Malaysia, and Hong Kong) that uses different currencies (SGD, MYR, HKD respectively). The raw data contains no FX conversion rates at the point of sale. Without converting the currencies, the revenue totals would be inaccurate, as the differing exchange rates would not be considered and all currencies would be assumed to hold equal value.

Because it is difficult to get the currency conversion rate on the day of the transaction, we used the average exchange rate for the past 5 years was used to convert the currencies all to SGD to solve this issue. For MYR, we applied a rate of 1 SGD = 3.3 MYR, and for HKD, a rate of 1 SGD = 6.1 HKD.

3.2 Zero Revenue and Zero Discount Orders

We found that 336 orders simultaneously record a value of S$0 in both the “Price: Total” = S$0 and “Price: Total Discount” = S$0 columns. These zero-value transactions are not isolated incidents, as they appear across multiple geographical regions, sales channels, and fiscal years. This overlapping pattern raises concerns regarding overall data accuracy. Because these records contain no financial values yet are logged as completed transactions, they compromises the integrity of sales transaction logs.

We judged that these orders were likely non-legitimate and were caused by a data import or system error, as they provide no meaningful value and would skew the total number of orders, rendering the analysis inaccurate. To mitigate this issue, we decided to remove them from our analysis.

3.3 100%-Discount Orders

A total of 1,281 orders record “Price: Total” = S$0 but a positive “Price: Total Discount”, indicating that the full product value has been discounted. We cross-referenced these transactions with Lush Protein and confirmed that these entries represent complimentary packages issued for marketing promotions, such as influencer collaborations. Because these items were fulfilled through the system, they appear as transactions with a 100% markdown rate.

These promotional orders shift our overall discount-rate metrics upward, but they do not affect the core retention or customer-lifetime-value calculations. These orders will be excluded from the final analysis because complimentary fulfilments do not reflect consumer purchase behaviour and should not influence conclusions regarding LTV, repeat rates, or customer loyalty.

3.4 Outlier / Wholesale Orders

We identified 78 B2B/outlier orders in total: 65 orders tagged wholesale-sale and 14 orders with Price: Total > S$5,000 (45 exceed S$5,000; one order meets both criteria, so 65 + 14 − 1 = 78 unique). These orders span multiple years and occur in the SG and MY regions. This small number of unusually large orders, including one explicitly tagged wholesale-sale in March 2026. Due to these distinct traits, we classified these high-value entries as corporate or B2B outliers that do not match standard retail profiles.

The presence of these large bulk orders introduces variance into our customer-lifetime-value calculations, making the affected customer segments appear uncharacteristically valuable. This distorts our baseline customer averages, though it represents a very small percentage of total order rows. To address this distortion, we revised the approach taken in our mid-term report, where these rows were temporarily retained. Now that Lush Protein has confirmed these are corporate accounts, we will exclude all 78 B2B/outlier orders (wholesale-sale tag OR Price: Total > S$5,000) from our finals analysis to ensure that our retail retention metrics remain reliable. The midterm EDA base (27,350 orders) retains them; the finals layer drops them.

3.5 Missing Values in UTM Attributions

Due to a significant tracking limitation, 94.9% of analysed orders lack essential UTM source, medium, and campaign data. This required the channel classification logic to rely on a fallback rule based on Shopify order tags and product handles. Although server-side tags reliably categorize Marketplace channels and identify Subscription orders with 99.9% consistency, the absence of UTM attribution impacts the remaining data. Because "Direct / Organic" serves as the default fallback category for any untracked traffic, any paid social or paid search campaigns lacking proper UTM tags are erroneously grouped here. This misclassification inflates the Direct/Organic customer count to 6,920 individuals and overstates the associated LTV and repeat-rate conclusions.

As the final LTV metrics for true organic traffic are likely skewed by this untracked paid acquisition, this limitation must be acknowledged. To mitigate this tracking error and account for the inherent attribution ambiguity, we decided to take following actions. First, Marketplace channels (Shopee and Lazada) are identified via Shopify tags, which are highly reliable as they are applied server-side by integrated apps during order creation. Second, we cross-referenced the Subscription orders using both the tags and the "is_subscription" flag, demonstrating an exceptional 99.9% consistency rate by matching in 3,263 out of 3,265 cases. Finally, to address the remaining attribution ambiguity surrounding untracked traffic, the Direct/Organic segment is transparently presented as 'Direct/Own channels' in our report and presentation.

3.6 Line-Item vs Order-Level Row Structure

The raw Shopify export contains a mixed row structure that combines order-header summary rows with individual order-line-item rows. Specifically, 13,484 rows (49.3% of the dataset) have a null value in the "Line: Product Handle" column. This indicates that nearly half of the rows function as order-level summaries and do not contain individual product data. Because the dataset lacks a uniform row structure, these two reporting levels must be separated to ensure accurate data aggregation.

Treating all 153,828 raw export rows as distinct orders would incorrectly inflate transaction volume and distort customer purchase history (the same order_id appears on both header and line rows). Furthermore, product-level analysis cannot use order-header rows that lack a line product handle. To resolve this, we built two tables in 01_load_and_merge.py: (1) order-level metrics using Top Row = 1 only - 27,350 orders; (2) line-level metrics using Line: Type = 'Line Item' - 50,963-line items. The customer table aggregates from the 27,350 order-level rows, not from the full raw row count.

3.7 Subscription Tag vs Flag Consistency

The dataset exhibits inconsistency between subscription order tags and the system's "is_subscription" boolean flag. Specifically, the data contains 3,267 orders tagged as subscription-related, compared to 3,265 orders where the "is_subscription" flag is set to true. This results in a minimal error margin where two specific orders possess the tracking tag but lack the corresponding boolean flag.

The operational impact of this discrepancy is negligible, as only 2 out of 3,267 subscription orders (0.06%) are misclassified. To mitigate this and maintain data uniformity, we decided to use "is_subscription" flag consistently across all subsequent data models.

3.8 Marketplace Subscription Tracking Gap and Repeat Rate Measurement

The Marketplace channel metrics are impacted by two related data limitations. First, a subscription tracking gap occurs because Shopify cannot ingest data from third-party platforms. Out of 3,268 marketplace orders, zero are flagged as subscriptions, resulting in a 0.0% subscription rate for all 2,126 marketplace customers. This represents a platform limitation rather than true consumer behaviour. Second, the marketplace repeat rate is restricted to Shopify-visible data. The 14.4% repeat rate is computed from 306 customers with two or more Shopify-visible orders divided by 2,126 marketplace-first customers. The 3,268 total marketplace orders confirm that repeat purchases are integrated, though undercounted due to unlinked customer profiles.

While the 0% subscription rate is a technical constraint of the Shopify dataset, the true external volume remains unknown. Similarly, the 14.4% repeat rate is likely a modest undercount. However, this does not invalidate the analysis; marketplace-exclusive repeat buyers offer no CRM visibility, reinforcing the argument that this channel yields lower-quality customer data. Furthermore, comparing the 14.4% Marketplace rate against the 33.5% Direct/Organic rate remains valid as both use the same baseline. To mitigate this, the final report will include annotations clarifying that marketplace platforms operate independent, untracked subscription systems, and that the repeat rate excludes transactions outside the Shopify ecosystem.

3.9 Unused Campaigns Dataset and Traffic Attribution Limits

We chose not to import or analyze the campaigns dataset "4_1. Sessions by referrer_20260505.csv" in our EDA. While it aggregates 246,909 total sessions and captures primary traffic drivers like Facebook and Meta, we identified critical tracking gaps. Specifically, we found no date columns, meaning five years of traffic data are collapsed into static totals. Furthermore, we discovered the file completely lacks customer and order IDs, making it impossible to link session behaviours to final purchases.

Because this file lacks timestamps and transaction IDs, we cannot perform time-trend analysis or compute true session-to-order conversion rates. To mitigate this, we utilized the order-level Browser: UTM Source field instead. This field contains the exact same marketing information but maintains full customer linkage. This approach allows our channel classification model to accurately map marketing touchpoints directly to actual transaction histories, proving far superior for lifetime value and customer behaviour analysis.

3.10 Discount Code Not Linked to Individual Orders

We found that the "3. Discounts" dataset tracks 367 discount codes and their total redemptions, but we couldn't link this data directly to individual order transactions because there was not any field that connects specific codes to single purchases from the raw Shopify export columns. The export only provides the discount amounts through the "Price: Total Discount" and "Line: Discount columns", leaving out the code names. Because of this missing data, we had to keep our discount taxonomy analysis in EDA as an isolated review of the "3. Discounts" table.

Because of this structural limitation, we could not identify which specific customers used which promotional campaigns. However, this issue does not change our findings on customer lifetime value, because we evaluate overall discount sensitivity using the binary "has discount" variable. To mitigate the missing code data, we adjusted our approach in EDA to categorize the "Price: Total Discount" amount as a percentage of the total order value. This method allows us to successfully verify our main finding that full-price buyers have a repeat purchase rate twice as high as buyers who receive a discount of 50% or more.

3.11 Temporal Coverage Gaps and Customer ID Mismatches in Recharge Data

We found that the Recharge subscription platform export only covers data from April 2025 to April 2026. However, Shopify order tags show that subscriptions have been running since at least 2021. Additionally, we discovered that Recharge uses an internal 8-digit customer ID that has zero matches with the 13-digit Shopify customer IDs. To bridge this gap, we joined the datasets using "shopify_order_id", which successfully matched 1,163 of the 1,215 Recharge rows. The remaining 52 unmatched rows represent a minor export date discrepancy in early April 2026. This ID mismatch explains why we see 1,095 unique subscription customers in Shopify but only 575 in Recharge.

Because of this data gap, our subscription churn analysis only represents the most recent year of data. The 526 recorded cancellation reasons reflect 2025–2026 behaviour and may not represent historical churn trends. However, our core retention and LTV metrics remain accurate because our comparison models utilize Shopify data tags, which captured all 1,095 subscription users. We address this constraint by clearly noting in our report that the churn findings represent a one-year window from the Recharge export. Additionally, we use the “shopify_order_id” field as our primary join key to maintain reliable cross-platform validation.

3.12 Product Master COGS Not Available

There was limitation in the “2.product_master file” where 65% of the listed variant rows contain null values in the Cost per item column. Out of 167 total variants - spanning 34 active, 22 draft, and 1 archived SKU - only 58 profiles contain active cost information. The lack of comprehensive records prevents us from conducting localized gross margin calculations at the SKU level. Additionally, the exact range of the available cost data cannot be verified in our analysis because the “Cost per item” field is classified as a sensitive business metric.

We could not calculate exact product costs so all profitability metrics in the report rely on an estimated 40% gross margin assumption. We applied a fixed assumption by setting MARGIN_RATE = 0.40 to approximate business value (profit figures are labelled "profit proxy") and stated in all attached script so the missing data does not affect our relative product comparisons and revenue-based findings.

4. Follow-Up Analysis to the Midterm Presentation

4.1 LTV (Life-time value) definition

4.1.1 Primary definition

LTV (customer) = Sum (Price: Total)

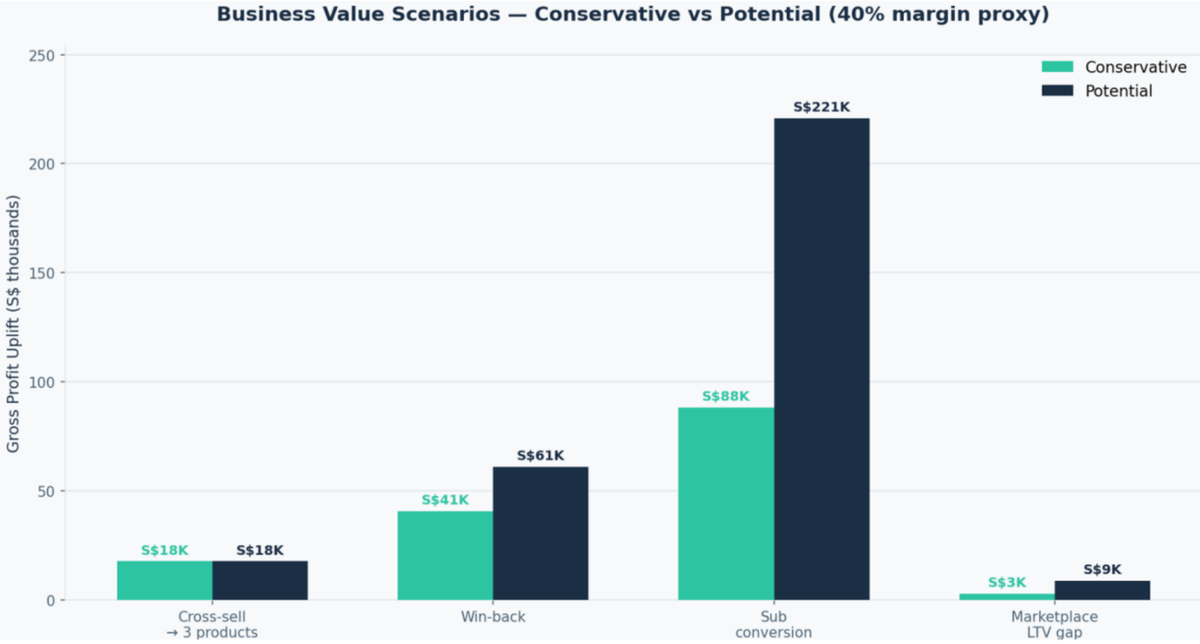
LTV per customer: Total revenue per customer across all paid, non-restocked orders for that customer_id, after FX conversion to SGD

4.1.2 Gross profit LTV (when margin needed)

Gross Profit LTV = LTV x margin rate

Margin rate = 0.40 (40% proxy – used because 65% of product master COGS is null)

4.2 Business Value Calculation (Mid-term presentation scenarios)



4.2.1 Cross-sell business value

Scenario	Pool	Rate	Uplift/customer	Gross LTV	Gross profit (40%)
2-product (conservative)	2,941	10%	S$60 (S$230–S$170)	**S$17,565**	**S$7,026**
3-product (conservative)	2,941	5%	S$300 (S$470–S$170)	**S$44,051**	**S$17,620**

4.2.2 Retention win-back (2024-25, no 2nd order)

Scenario	Pool	Rate	Value/customer	Gross LTV	Gross profit
Conservative	2,346	10%	S$336*	**S$78,741**	**S$31,496**
Potential	2,346	15%	S$336*	**S$118,112**	**S$47,245**

4.2.3 Subscription conversion (Discount strategy)

Scenario	Pool (non-subs)	Rate	Uplift (S$532–S$200)	Gross LTV	Gross profit
Conservative	12,685	5%	S$332	**S$210,603**	**S$84,241**
Potential	12,685	12.5%	S$332	**S$526,507**	**S$210,603**

4.2.4 Marketplace → direct LTV gap

Scenario	Pool	Rate	Gap (Direct–Marketplace LTV)	Gross LTV	Gross profit
Conservative	2,126	5%	S$83	**S$8,806**	**S$3,523**
Potential	2,126	15%	S$83	**S$26,419**	**S$10,568**

4.2.5 What changed from mid-term presentation (PPT)

Midterm slide	**Finals adjustment**
K$528 subscription upside	Still valid at 12.5% non-sub conversion; now split into conservative (5%) vs potential (12.5%)
K$219 retention	Recalculated on filtered pool: 2,346 × 10–15% × S$336
51%+ discount story	**Removed from finals narrative** per LushProtein — still included in historical EDA, excluded from eligible cohort
S$190 vs S$198 website LTV	Unchanged

5. Applied Analysis Filters – LushProtein Feedback

Following the midterm presentation, LushProtein provided additional guidance on which customer segments should be included in the finals analysis. Note that these are not data quality issues. The objective was to focus the analysis on customers that best represent the company's current target market and business model.

Four filters were applied. First, customers who purchased the *Better Whey Protein Elite* product were excluded. LushProtein indicated that these customers tend to purchase in bulk and do not represent the purchasing behaviour of the broader customer base.

Second, customers acquired in July and November were excluded. These months contain major promotional campaigns, including the company's anniversary promotions and Black Friday sales. Customers acquired during these periods are generally more promotion-driven and may not reflect typical purchasing behaviour.

Third, only customers acquired from January 2022 onwards were included. According to LushProtein, the years before 2022 were a period of product and pricing experimentation. Restricting the analysis to more recent customers ensures that the findings are based on the company's current product offering and operating model.

Lastly, customers whose first purchase received a discount of more than 50% were excluded. These customers were typically acquired through referral programmes, sampling campaigns, or other heavily subsidised promotions. As the focus of this project is to understand long-term customer behaviour, including these customers could bias the retention and lifetime value analysis.

The final customer pool used for the analysis is summarized in below table.

Cohort Filtering Stage	**Customers Remaining**	**Notes**
Total customer base	13,780	All paid, non-restocked customers in orders.parquet
Acquired on or after 1 January 2022	~6,200	Removes pre-2022 experimentation-era acquisitions (~7,500 customers)
Additional business-rule filters applied*	6,353	Finals-eligible pool; each customer counted once
Final finals-eligible cohort	**6,353**	

Filters are applied together in the finals script; customers meeting any exclusion rule are removed once.